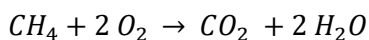
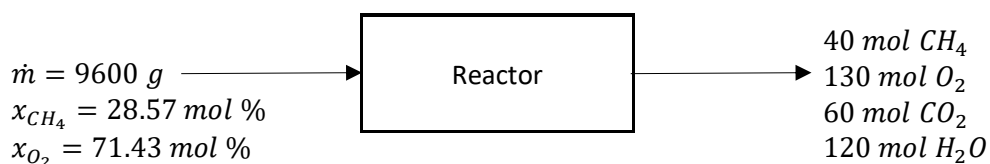


Use the information below to answer questions (1) through (3). Methane is burned to form carbon dioxide and water in a batch reactor:

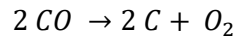


The feed to the reactor and the products obtained are shown in the following flowchart:



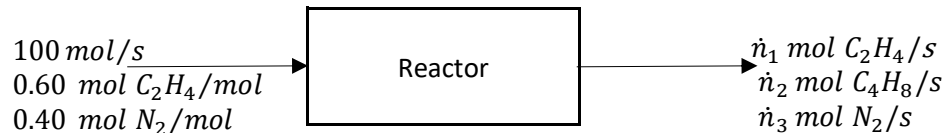
- 1) What is the fractional conversion of methane?
 - a. 0.9958
 - b. 0.9854
 - c. 0.7998
 - d. 0.5998
- 2) What is the extent of the reaction for the reaction equation above?
 - a. -120 mol
 - b. -60 mol
 - c. 60 mol
 - d. 120 mol
- 3) It was decided that air is to be used to supply oxygen in the above batch reactor; but it was found that an excess of 30% air is also needed to prevent side reactions. Assume the same amount of methane is fed to the reactor and air composition of 21 mol% O₂ / 79 mol% N₂. Determine the mass of air required to meet this new criterion.
 - a. 17.9 kg
 - b. 35.7 kg
 - c. 44.7 kg
 - d. 74.3 kg
- 4) Suppose the formula $\Delta \hat{H} = \int_{T_1}^{T_2} C_p(T) dt$ is used to calculate the specific enthalpy change for a change in temperature and pressure undergone by: an ideal gas, a highly non-ideal gas, and a liquid. Rank these for which the formula will estimate the specific enthalpy most accurately.
 - a. ideal gas > liquid > highly non-ideal gas
 - b. liquid > ideal gas > highly non-ideal gas
 - c. ideal gas > highly non-ideal gas > liquid
 - d. ideal gas = liquid > highly non-ideal gas

- 5) The standard heat of the reaction



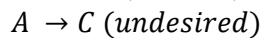
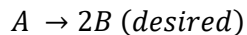
is $\Delta \hat{H}_r^\circ = +221.0 \frac{\text{kJ}}{\text{mol}}$. What is the standard heat of formation of CO?

- 221.0 kJ/mol
 - 110.5 kJ/mol**
 - 110.5 kJ/mol
 - Not enough information.
- 6) If substance A and substance B each have a specific gravity of 1.34, must 3 cm³ of A have the same mass as 3 cm³ of B?
- Yes
 - No**
- 7) A mixture of ethylene and nitrogen is fed to a reactor in which some of the ethylene is dimerized to butene.



How many independent atomic species balances are in the process above?

- 0
 - 1
 - 2**
 - 3
- Use the information below to answer questions (8) through (10). Consider the following pair of reactions:



Suppose 100 mol of A is fed to a batch reactor and the final product contains 10 mol of A, 160 mol of B, and 10 mol of C.

- What is the percentage yield of B?
 - 20%
 - 40%
 - 60%
 - 80%**
- What is the selectivity of B relative to C?
 - 16**
 - 8
 - 4
 - 0.0625
- What are the extents of the first (desired) and second (undesired) reactions?
 - $\xi_1 = 160 \text{ mol}, \xi_2 = 10 \text{ mol}$
 - $\xi_1 = -80 \text{ mol}, \xi_2 = -10 \text{ mol}$
 - $\xi_1 = 80 \text{ mol}, \xi_2 = 10 \text{ mol}$**
 - $\xi_1 = -160 \text{ mol}, \xi_2 = -10 \text{ mol}$

Homework Problem 6.48 (Murphy)

A waste gas stream contains 40 mol% ethane (C_2H_6), 19 mol% N_2 , 1 mol% H_2S , 10 mol% O_2 , and 30 mol% CO . The gas is fed to a furnace along with 20% excess air and combusted. The waste stream entering the furnace is at 25°C . The air enters the furnace at 25°C . The flue gas exits the furnace stack at 150°C . Complete combustion can be assumed. The heat generated in the furnace is used to generate saturated steam at 10 bar, using boiler feedwater at 50°C and 10 bar. Calculate the amount of steam generated in the furnace per gmol waste gas fed (kg steam per gmol gas).

